UNIT - 5

* **GCD :**

Let A be a no and b be +ve integer and have unique quotient and remainder such that $a = bq + r$ where $0 \leq r < b$.

* **Divisibility:** Suppose a and b be any 2 integers where $a \neq 0$, then a divides b if there exists an integer k such that $b = ak$ and is denoted by a/b. It is also called as a is a factor of b or b is divisible by a or b is multiple of a.

Eg: $3/12$, $2/6$, $4/8$, $2 \not| 3$, $2 \not| 1$

NOTE: Quotient q maybe +ve, -ve or 0 but remainder r is always +ve (or 0) and less than divisor.

* **Proper divisor:** A proper divisor of a +ve integer A are those nos other than a itself that divide a without remainder.

Eg: $6 \rightarrow 1, 2, 3$
$12 \rightarrow 1, 2, 3, 4, 6$

* **Greatest Common Divisor:** The GCD of 2 integers a and b (both $a \neq 0 \neq b$) is a unique +ve integer d such that
  - d/a and d/b.
  - If c/a and c/b then c/d.

Eg: $6 \rightarrow 1, 2, 3, 6$ $\qquad c \rightarrow 2, 3$
$12 \rightarrow 1, 2, 3, 4, 6, 12.$ $\qquad d \rightarrow 6.$

It is denoted by $\gcd(a,b) = d$ or $(a,b) = d$.

NOTE: • If $\gcd(a,b) = 1$ then 2 integers a and b are said to be relatively prime.
eg: $(3,4) = 1$, $(3,7) = 1$

• If $\gcd(a,b) = d$ then there exists r and s such that $ar + bs = d$.

* Let a, b, c, m, n be any integer then:

(a) If a/b & b/c then a/c.
(b) If a/b & a/c then $a/(bm+nc)$
(c) If a/bc & $\gcd(a,b) = 1$ then a/c.
(d) If a/c, b/c and $(a,b) = 1$ then ab/c.
(e) If $\gcd(a,b) = 1 = (a,c)$ then $\gcd(a,bc) = 1$
(f) If $\gcd(a,b) = 1$ & c/a, d/b then $(c,d) = 1$

(a) By divisibility definition, if
If a/b then $b = ak_1 \longrightarrow (1)$, $k_1 \in \mathbb{Z}$
If b/c then $c = bk_2 \longrightarrow (2)$ where $k_2 \in \mathbb{Z}$

$\Rightarrow c = ak_1k_2$
$c = ak_3$ where $k_3 \in \mathbb{Z}$
$\Rightarrow a/c$.

(b) If a/b then $b = ak_1$ where $k_1 \in \mathbb{Z}$
If a/c then $c = ak_2$ where $k_2 \in \mathbb{Z}$
Consider, $bm + nc = ak_1m + \cancel{ak_2}\, ak_2n$
$= a\underbrace{(k_1m + k_2n)}$ → integer.
$\Rightarrow bm + nc = ak_3$ where $k_3 = k_1m + k_2n$.
$\Rightarrow a/(bm+nc)$

a divides b
$b = ak$

15/20 20/5

(c) $(a,b) = 1$
there exists $r, s \in \mathbb{Z} \Rightarrow ar + bs = 1$
$\Rightarrow acr + bcs = c$

If $a | bc$ then $a | bcs$.
If $a | a$ then $a | acr$

from (b) $\Rightarrow a | (bcs + acr) = a | c$.

(d). $a | c$ then $c = ak_1$, $k_1 \in \mathbb{Z}$
$b | c$ then $c = bk_2$, $k_2 \in \mathbb{Z}$

$(a,b) = 1$ then $ar + bs = 1$, $r, s \in \mathbb{Z}$
$\Rightarrow acr + bcs = c$
$\Rightarrow abk_2 r + bak_1 s = c$
$\Rightarrow ab(k_2 r + k_1 s) = c$
$\Rightarrow abk_3 = c$ where $k_3 = k_2 r + k_1 s$.
$\Rightarrow$ ~~c/ab~~ $ab | c$

e) (a,b)
There exist $r_1, s_1 \in \mathbb{Z}$ where $ar_1 + bs_1 = 1 \Rightarrow bs_1 = 1 - ar_1$
" " r_2, s_2 " " $ar_2 + cs_2 = 1 \Rightarrow cs_2 = 1 - ar_2$

$\Rightarrow bcs_1 s_2 = 1 - ar_2 - ar_1 + a^2 r_1 r_2$

$\Rightarrow bcs_1 s_2 = 1 - a(r_1 + r_2 - ar_1 r_2)$
$s = s_1 s_2$, $r = r_1 + r_2 - ar_1 r_2$
$\Rightarrow bcs = 1 - ar$
$\Rightarrow sbc + ar = 1 \Rightarrow (a, bc) = 1$.

f) $ar + bs = 1$

$a = ck \quad , \quad b = dk$.

$ck\cancel{s_1} + dks = 1$.

$ck_1 + dk_2 = 1 \Rightarrow (c,d) = 1$

* **Properties of GCD:**

1. $(a,b) = (b,a)$
2. $(a,b) = (a,-b) = (-a,b) = (-a,-b)$
3. If b/a then $(a,b) = b$.
4. $(0,a) = a$.
5. $(0,0) =$ not defined.

* **Euclid's Algorithm:** Let a, b be 2 +ve integers s.t $(b < a)$ and dividing a by b, we get $a = bq_1 + r_1$ where $0 \leq r_1 < b$. If:

$\rightarrow r_1 = 0$: $a = bq_1 \Rightarrow b/a$.

$\Rightarrow (a,b) = b$.

$\rightarrow r_1 \neq 0$: $(a = bq_1 + r_1)$ Any common divisor of a, b is divisor of r_1

$\Rightarrow (a,b) = (b, r_1)$

Again dividing b by r_1 we get $b = r_1 q_2 + r_2$ where $0 < r_2 < r_1$. Here $(b, r_1) = (r_1, r_2)$

$||^{ly}$ we get $r_1 (= r_2 q_3 + r_3), r_2, \ldots r_n$ which is decreasing sequence of nos. and is continued till remainder equals 0.

~~Suppose~~

$\Rightarrow r_{n-1} = r_n q_{n+1} + r_{n+1}$.

If $r_{n+1} = 0$: $r_{n-1} = r_n q_{n+1}$
$\Rightarrow r_n / r_{n-1}$

$\Rightarrow (r_n, r_{n-1}) = r_n$

$(a,b) = (a, r_1) = (r_1, r_2) = (r_{n-1}, r_n) = r_n$
At last non-zero divisor is the gcd of a, b.

$\Rightarrow (a,b) = r_n$.

13)* Find gcd of following using division algorithm:

1) 27, 87 $\Rightarrow$ $87 = 27(3) + 6$
$\Rightarrow 27 = 6(4) + 3$
$\Rightarrow 6 = 3(2) + 0$.

$\Rightarrow$ gcd $(27, 87) = \underline{\underline{3}}$.

2) 165, 418 $\Rightarrow$ $418 = 165(2) + 88$
$165 = 88(1) + 77$.
$88 = 77(1) + 11$.
$77 = 11(7) + 0$.
$\Rightarrow$ gcd $(165, 418) = \underline{\underline{11}}$.

3) 30, 42, 70 $\Rightarrow$ $42 = 30(1) + 12$
$30 = 12(2) + 6$
$12 = 6(2) + 0$.
gcd $(30, 42) = 6$.

$70 = 6(11) + 4$.
$6 = 4(1) + 2$
$4 = 2(2) + 0$. $\Rightarrow$ $(6, 70) = 2$.
$\Rightarrow (30, 42, 70) = 2$.

1652.

14) $1890 = 826(2) + \cancel{238}\ 238$

$826 = 238(3) + 112$

$238 = 112(2) + 14$

$112 = 14(8) + 0 \Rightarrow d = 14 = (1890, 826)$

$14 = (1)238 + 112(-2) \Rightarrow (238, 112) = 14$

$14 = 238(1) + (826 - 238(3))(-2)$

$14 = 238(7) + 826(-2) \Rightarrow (238, 826) = 14$

$\cancel{14 = 238(7) + (1890 - 826(2))}$

$14 = 826(-2) + (1890 - 826(2))(7)$

$14 = \underline{826(-16)} + 1890(7) \Rightarrow (826, 1890) = 14$

16) $(27, 87) = 3$

$3 = 27(1) + 6(-4)$

$3 = 27(1) + (87 + 27(-3))(-4)$

$3 = (27)(13) + (87)(-4)$

* <u>Linear Diophantine Eqn</u>: An eqn of the $ax + by = c$ where $a \neq 0 \neq b$ and c is an integer is called a Linear Diophantine Eqn in 2 variables x, y.

* <u>Soln of Linear Diophantine Eqn (LDE)</u>:

1) If $\gcd(a, b) = d$ then LDE $ax + by = c$ has an integral soln only when d/c.
2) ~~If~~ If (x_0, y_0) are pair of integers is called a particular soln of LDE when $ax_0 + by_0 = c$
3) The general soln of LDE is given by:

$$x = x_0 + \frac{bt}{d} \quad ; \quad y = y_0 - \frac{at}{d}$$

$$x = x_0 + \left[\frac{\text{coeff of } y}{\gcd(a,b)}\right] t \quad \text{where } t \in \mathbb{Z}$$

$$y = y_0 \mp \left[\frac{\text{coeff of } x}{\gcd(a,b)}\right] t$$

19) i) $a=12,\ b=18,\ c=30$

$18 = 12(1) + 6$
$12 = 6(2) + 0 \Rightarrow \gcd(18,12) = 6 = d$
$6/30 = d/c \Rightarrow$ Solution exists

ii) $a=2,\ b=3,\ c=4$
$3 = 2(1) + 1$
$2 = 1(2) + 0 \Rightarrow \gcd(3,2) = 1 = d$
$1/4 \Rightarrow d/c \Rightarrow$ Solution ~~does not~~ exist

20) i) $28x + 91y = 119$
$a = 28,\ b = 91,\ c = 119$
$91 = 28(3) + 7$
$28 = 7(4) + 0 \Rightarrow (28, 91) = 7$
$7/119 \Rightarrow$ Soln exists

$7 = 91 - 28(3) \Rightarrow 7\times 17 = 91(17) - 28(3)(17)$
$\Rightarrow 119 = 91(17) + 28(-51)$
Particular soln $\Rightarrow x_0 = -51,\ y_0 = 17$
General soln $\Rightarrow x = -51 + \frac{91}{7}(t)$

$x = 13t - 51$
$y = 17 - 4t$

II $\left[\begin{array}{c|cc} a & 1 & 0 \\ b & 0 & 1 \end{array}\right]$

By row reduction $\Rightarrow \left[\begin{array}{c|cc} d & s & t \\ 0 & s_1 & t_1 \end{array}\right]$

Case-1: $d=c$:

$x = s + ks_1$

$y = t + kt_1$

Case-2: $d \neq c$:

$d/c \Rightarrow c = c'd$

$x = sc' + ks_1$

$y = tc' + kt_1$

$\left[\begin{array}{c|cc} 28 & 1 & 0 \\ 91 & 0 & 1 \end{array}\right] \Rightarrow R_2 \to R_2 - 3R_1$

$\Rightarrow \left[\begin{array}{c|cc} 28 & 1 & 0 \\ 7 & -3 & 1 \end{array}\right]$.

$R_1 \to R_1 - 4R_2$

$\Rightarrow \left[\begin{array}{c|cc} 0 & 13 & -4 \\ 7 & -3 & 1 \end{array}\right] \Rightarrow \left[\begin{array}{c|cc} 7 & -3 & 1 \\ 0 & 13 & -4 \end{array}\right]$

$7/119 \Rightarrow 119 = 7 \times 17$

$\Rightarrow c' = 17$

$x = -3(17) + k(13) = 13k - 51$

$y = 1(17) + k(-4) = 17 - 4k$.

20) iii) $15x + 21y = 39$.

$a = 15 \quad b = 21, \quad c = 39$

$21 = 15(1) + 6$

$15 = 6(2) + 3$

$6 = 3(2) + 0 \Rightarrow (15, 21) = 3$.

$3/39 \Rightarrow$ Soln exists

63
46
17

$3 = 15 - 6(2)$
$3 = 15 - (2)(21-15)$
$3 = 15(3) + 21(-2)$
$\Rightarrow 3(13) = 15(3)(13) + 21(-2)(13)$
$\Rightarrow 39 = 15(39) + 21(-26)$

$x_0 = 39,\ y_0 = -26 \rightarrow$ Particular soln.

$$\left.\begin{aligned} x &= x_0 + \frac{b}{d}t = 39 + 7t \\ y &= y_0 - \frac{a}{d}t = -26 - 5t \end{aligned}\right\}$$ General soln.

20)(ii) $63x - 23y = -7$
$a = 63,\ b = -23,\ c = -7$

$63 = 23(2) + 17$
$23 = 17(1) + 6$
$17 = 6(2) + 5$
$6 = 5(1) + 1$
$5 = 5(1) + 0 \Rightarrow (63, -23) = 1 = d.$
$1 \mid -7 \Rightarrow$ Soln exists.

$\cancel{1 = 7 - 2(3)}$
$\cancel{1 = 7 - 3(16 - 7(2))}$
$\cancel{1 = 7(3) + 16(-3)}$

$1 = 6 - 5(1)$
$1 = 6 - (17 - 6(2))$
$1 = 6(3) - 17(1)$
$1 = -17(1) + 3(23 - 17)$
$1 = 17(-4) + 23(3)$
$1 = 23(3) + (63 - 23(2))(-4)$
$1 = 23(11) + 63(-4)$
$-7 = -23(77) + 63(28)$
$x_0 = 28,\ y_0 = 77$

$x = 28 - 23t$, $y = 77 - 63t$

$$\left[\begin{array}{c|cc} 63 & 1 & 0 \\ -23 & 0 & 1 \end{array}\right] \Rightarrow \left[\begin{array}{c|cc} 17 & 1 & 2 \\ -23 & 0 & 1 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{c|cc} 17 & 1 & 2 \\ -6 & 1 & 3 \end{array}\right] \Rightarrow \left[\begin{array}{c|cc} 5 & 3 & 11 \\ -6 & 1 & 3 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{c|cc} 5 & 3 & 11 \\ -1 & 4 & 14 \end{array}\right] \Rightarrow \left[\begin{array}{c|cc} 0 & 23 & 81 \\ -1 & 4 & 14 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{c|cc} -1 & 4 & 14 \\ 0 & 23 & 81 \end{array}\right] \Rightarrow$$

* Congruency:

Let m be any +ve integer $(m \geq 1)$

If a and b be 2 integers, then a is said to be congruent to b modulo m. if and only if m divides a-b and it is denoted as $a \equiv b \pmod m$.

where $\equiv$ is a symbol of congruence.

$[m / a-b \Rightarrow a \equiv b \pmod m]$

Eg: $25 \equiv 3 \pmod{11} \Rightarrow a = 25, b = 3, m = 11$

$\Rightarrow m / a-b = 11/25-3 = 11/22$ ✓

$79 \equiv 8 \pmod 9 \Rightarrow 9 / 79-8 = 9 \nmid 71$

$\Rightarrow 79 \not\equiv 8 \pmod 9$.

NOTE: If m does not divide a-b then a is not congruent to b modulo(m) and is denoted by $a \not\equiv b \pmod m$

* Properties of congruency:

1) If $a \equiv b \pmod{m}$ then a is expressed as:
$a = b + mk$ where $k \in \mathbb{Z}$

2) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then:

$$a \pm c \equiv b \pm d \pmod{m}$$
$$ac \equiv bd \pmod{m}$$

3) $a^n \equiv b^n \pmod{m}$ where $n \in \mathbb{Z}^+$

4) If and only if $a \equiv b \pmod{m}$, a, b leaves the same remainder when divided by m

5) If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$ then $a \equiv b \pmod{m}$

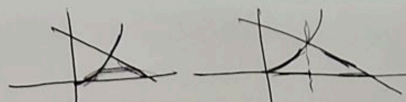
6) If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = d$ then $a \equiv b \left(\bmod \left(\frac{m}{d}\right)\right)$

* Linear Congruency: If a, b, m are integers s.t $a \not\equiv 0 \pmod{m}$ and x is unknown then the congruence $ax \equiv b \pmod{m}$ is called a linear congruency.

→ Suppose x_0 is the soln of the congruence then $ax_0 \equiv b \pmod{m}$

→ If x_0 is congruent to $x_1 (\equiv x_0 \pmod{m})$ then x_1 is also the soln of $ax \equiv b \pmod{m}$ i.e if x_0 is a soln then every member of its congruence class is also a soln

$27x \equiv 3 \pmod{11}$



→ If $\gcd(a,m)=d$ and d divides b then the linear congruence has exactly d no. of soln in the set $\{0,1,2,\dots m-1\}$ and are called as incongruent solns and it is given by x,

$x = x_0 + \left(\frac{m}{d}\right)t$ where $0 \le t < d$.

NOTE: $x = x_0 + mk$, $(k \in \mathbb{Z})$ is general soln.

→ The linear congruence $ax \equiv b \pmod m$ has unique soln if & only if $\gcd(a,m)=1$.

* $5x \equiv 4 \pmod{13}$

$a=5$, $b=4$, $m=13$.

$(a,m) = (5,13) = 1$

Clearly $1/4 \Rightarrow d/b$

$\Rightarrow$ Soln exists

$\Rightarrow$ d gives the no. of incongruent solns

$\Rightarrow d=1 \Rightarrow 1$ incongruent soln in $\{0,1,\dots 12\}$

$13/5x-4 \Rightarrow 5x-4 = 13k$

$x = \frac{13k+4}{5}$

By inspection, $k=2$, $x=6$

$\Rightarrow x \equiv 6 \pmod{13}$ is the soln.

General soln is $x = 6+13t \ \forall t \in \mathbb{Z}$.

* $7x \equiv 9 \pmod{15}$

$a=7$, $b=9$, $c=15$

$(a,m) = (7,15) = 1$

Clearly $1/9 \Rightarrow d/b \Rightarrow$ Soln exists.

$\Rightarrow d=1 =$ incongruent soln.

$15/7x-9 \Rightarrow x = \frac{15k+9}{7} \Rightarrow k=5, x=12$.

$\Rightarrow x \equiv 12 \pmod{15} \longrightarrow x = 12+15t \ \forall t \in \mathbb{Z}$

If $a, b > m$ then divide a by m and write the remainder in the place of a.
IIly divide b by m and write the remainder in place of b.

30) i) $a = 27,\ b = 11,\ m = 7.$

$\Rightarrow 6x \equiv 4 \pmod 7$

$\Rightarrow 7 / 6x - 4 \Rightarrow x = \frac{7k+4}{6}$

$k = 2,\ x = 3$

$\Rightarrow x \equiv 3 \pmod 7$ is the soln.

General soln, $x = 3 + 7t.\ \forall t \in \mathbb{Z}$

(ii) $37x \equiv 5 \pmod{11}$

$\Rightarrow 4x \equiv 5 \pmod{11}$

$\Rightarrow 1/5 \Rightarrow 1$ incongruent soln.

$\Rightarrow 11 / 4x - 5 \Rightarrow x = \frac{11k+5}{4}$

$\Rightarrow k = 1,\ x = 4.$

$\Rightarrow x \equiv 4 \pmod{11}$

$\Rightarrow x = 4 + 11t.$

(iii) $15x \equiv 12 \pmod{36}.$

$a = 15,\ b = 12,\ m = 36.$

$(15, 36) = 3/12$

$\Rightarrow$ Soln exists. $\Rightarrow$ 3 incongruent soln in set $\{0, 1, \ldots 35\}$

$\Rightarrow$ gcd $= 3$

$\Rightarrow 5x \equiv 4 \pmod{12} \Rightarrow 12 / 5x - 4$

$5x - 4 = 12k \Rightarrow x = \frac{12k+4}{5}$

$\Rightarrow k=3,\ x=8$

$\therefore$ Incongruent soln are : $x = x_0 + \frac{mt}{d}$; $0 \le t < d$

$x = 8 + \frac{36t}{3}$

$\Rightarrow x = 8 + 12t$

$t=0,\ x=8$; $t=1,\ x=20$; $t=2,\ x=32$

$x = 8 + 12t$

$x = 20 + 12t$

$x = 32 + 12t$

(iv) $36x \equiv 96 \pmod{156}$

$(36, 156) = 12$

$12 / 96 \Rightarrow 12$ incongruent solns

$\Rightarrow 3x \equiv 8 \pmod{13}$

$\Rightarrow 13 / 3x-8 \Rightarrow x = \frac{13k+8}{3}$

$\Rightarrow k=1,\ x=7$

Incongruent solns are: $x = x_0 + \frac{m}{d}t$

$\Rightarrow x = 7 + \frac{156}{12}t \Rightarrow x = 7 + 13t$

$t=0,\ x=7 : x = 7 + 13t$

$t=1,\ x=20 : x = 20 + 13t$

$t=2,\ x=33 : x = 33 + 13t$

$t=3,\ x=46 : x = 46 + 13t$

$t=4,\ x=59 : x = 59 + 13t$

$t=5,\ x=72 : x = 72 + 13t$

$t=6,\ x=85 : x = 85 + 13t$

$t=7,\ x=98 : x = 98 + 13t$

$t=8,\ x=111 : x = 111 + 13t$

$t=9,\ x=124 : x = 124 + 13t$

$t=10,\ x=137 : x = 137 + 13t$

$t=11,\ x=150 : x = 150 + 13t$

* Relation bt Linear Diophantine Eqn and Congruence

A congruent eqn $ax \equiv b \pmod{m}$ can be written as $ax + my = b$

* Procedure to solve Linear Congruence:

1) For given $ax \equiv b \pmod{m}$ check whether $\gcd(a,m) \mid b$. If not then there is no soln otherwise proceed to (2).
2) Find particular soln x_0 to the given congruence by converting it into LDE $ax + my = b$.
3) Find general soln for x: $x = x_0 + \frac{mt}{d}$.
4) Then all the incongruent soln are numbers greater than 0, less than m.

eg: $15x \equiv 12 \pmod{36}$

$a = 15, b = 12, m = 36.$

$\Rightarrow 15x + 36y = 12.$

$(a,m) = (15,36) = 3.$

$36 = 15 \times 2 + 6.$

$15 = 2 \times 6 + 3.$

$6 = 3 \times 2 + 0.$

$\Rightarrow 3 = 15 + 6(-2)$

$\Rightarrow 3 = 15 + (36 - 15(2))(-2)$

$\Rightarrow 3 = 15(5) + 36(-2)$

$12 = 15(20) + 36(-8)$

$\Rightarrow x_0 = 20$ is particular soln.

$\Rightarrow x = 20 + 12t \quad \forall t \in \mathbb{Z}$

Solns are $\{\ldots -4, \boxed{8, 20, 32}, 44, \ldots\}$.

Incongruent soln are $x = 8, 20, 32$.

32) (i) $8x \equiv 10 \pmod 6$

$(8,6) = 2 \mid 10 \Rightarrow$ Soln exists

$8 = 6 \times 1 + 2 \qquad \Rightarrow 8x + 6y = 10$

$6 = 2 \times 1 + 0$

$\Rightarrow 2 = 8 + 6(-1)$

$\Rightarrow 10 = 8(5) + 6(-5)$

$\Rightarrow x_0 = 5$

$x = 5 + \frac{6}{2}t \Rightarrow x = 5 + 3t \quad \forall t \in \mathbb{Z}$

Solns are $\{\ldots -1, \boxed{2, 5}, 8, 11, \ldots\}$

$\Rightarrow$ Incongruent solns are $x = 2, 5$

(ii) $2x \equiv 3 \pmod 4$

$(2,4) = 2 \nmid 3 \Rightarrow$ Soln doesnt exist

(iii) $4x \equiv 7 \pmod 5$

$(4,5) = 1 \mid 7 \Rightarrow$ Soln exists

$\Rightarrow 4x + 5y = 7$

$\Rightarrow 5 = 4 \times 1 + 1 \qquad \Rightarrow 1 = 5 - 4$

$4 = 1 \times 4 + 0 \qquad \Rightarrow 7 = 5(7) + 4(-7)$

$\Rightarrow x_0 = -7 \Rightarrow x = -7 + 5t \quad \forall t \in \mathbb{Z}$

$\Rightarrow$ Solns are $\{\ldots -12, -7, -2, \boxed{3}, 8, \ldots\}$.

$\Rightarrow$ Incongruent solns are $x = 3$.

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* Chinese Remainder Theorem:

Let $m_1, m_2, \dots m_n$ be positive integers s.t $\gcd(m_i, m_j) = 1$ for $i \neq j$ m_i's / m_j's are relatively prime, then the system of linear congruences $x \equiv a_1 \pmod{m_1}$, $x \equiv a_2 \pmod{m_2}$ ... $x \equiv a_n \pmod{m_n}$ has a unique soln modulo to $M = m_1 m_2 \dots m_n$

Soln to the above system of congruence is given by

$$x \equiv (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_n M_n M_n^{-1}) \pmod M$$

where $M_i = \dfrac{M}{m_i}$

33) i) $x = 1 \pmod 3$
$x = 2 \pmod 5$
$x = 3 \pmod 7$

Given			To find.		
$a_1 = 1$	$m_1 = 3$		$M_1 = \frac{M}{m_1} = 35$	$M_1^{-1} = 2$	$a_1 M_1 M_1^{-1} = 70$
$a_2 = 2$	$m_2 = 5$	$M = m_1 m_2 m_3$ $M = 105$	$M_2 = \frac{M}{m_2} = 21$	$M_2^{-1} = 1$	$a_2 M_2 M_2^{-1} = 42$
$a_3 = 3$	$m_3 = 7$		$M_3 = \frac{M}{m_3} = 15$	$M_3^{-1} = 1$	$a_3 M_3 M_3^{-1} = 45$

$M_1 M_1^{-1} \equiv 1 \pmod 3$
$35(M_1^{-1}) \equiv 1 \pmod 3$
$35 \times 2 - 1 = 69$
$3 / 69 \Rightarrow M_1^{-1} = 2$

$M_2 M_2^{-1} \equiv 1 \pmod 5$
$21 M_2^{-1} \equiv 1 \pmod 5$
$21 \times 1 - 1 = 20$
$5/20 \Rightarrow M_2^{-1} = 1$

$M_3 M_3^{-1} \equiv 1 \pmod 7$.
$15 M_3^{-1} \equiv 1 \pmod 7$.
$M_3^{-1} = 1$
$\Rightarrow 15 \times 1 - 1 = 14$
$7 | 14 \Rightarrow M_3^{-1} = 1$.

Soln is : $X \equiv (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + a_3 M_3 M_3^{-1}) \pmod M$
$X \equiv (70 + 42 + 45) \pmod{105}$
$X \equiv 157 \pmod{105}$ → Use 157/105's remainder
$\underbrace{X \equiv 52 \pmod{105}}_{\text{least value}}$.
$\Rightarrow X = 52 + 105t \quad \forall\ t \in \mathbb{Z}$

NOTE: $M_1 M_1^{-1} \equiv 1 \pmod{m_1}$
$M_n M_n^{-1} \equiv 1 \pmod{m_n}$

33)(ii) $x = 2 \pmod 5$
$x = 3 \pmod 7$
$x = 4 \pmod 9$

Given			To find.		
$a_1 = 2$	$m_1 = 5$	$M = 315$	$M_1 = 63$	$M_1^{-1} = 2$	$a_1 M_1 M_1^{-1} = 252$
$a_2 = 3$	$m_2 = 7$		$M_2 = 45$	$M_2^{-1} = 5$	$a_2 M_2 M_2^{-1} = 675$
$a_3 = 4$	$m_3 = 9$		$M_3 = 35$	$M_3^{-1} = 8$	$a_3 M_3 M_3^{-1} = 1120$

$63 M_1^{-1} \equiv 1 \pmod 5$. $45 M_2^{-1} \equiv 1 \pmod 7$.
$M_1^{-1} = 2$. $M_2^{-1} = 5$
$63 \times 2 - 1 = 125$
$5 | 125$

$35 M_3^{-1} \equiv 1 \pmod 9$
$M_3^{-1} = 8$.

$X \equiv (252 + 675 + 1120) \pmod{315}$
$X \equiv 2047 \pmod{315}$
$X \equiv 157 \pmod{315}$

(iii) $x \equiv 2 \pmod 1$
$x \equiv 3 \pmod 2$
$x \equiv 4 \pmod 3$

$a_1 = 2$	$m_1 = 1$		$M_1 = 6$	$M_1^{-1} = 1$	12
$a_2 = 3$	$m_2 = 2$	$M = 6$	$M_2 = 3$	$M_2^{-1} = 1$	9
$a_3 = 4$	$m_3 = 3$		$M_3 = 2$	$M_3^{-1} = 2$	16

$M_1^{-1} \equiv 1 \pmod 1$
$M_1^{-1} = 1$

$3M_2^{-1} \equiv 1 \pmod 2$
$M_2^{-1} = 1$

$2M_3^{-1} \equiv 1 \pmod 3$
$M_3^{-1} = 2$

$X \equiv 37 \pmod 6$
$X \equiv 1 \pmod 6$

(iv) $x \equiv 2 \pmod 3$
$x \equiv 3 \pmod 7$
$x \equiv 5 \pmod{11}$

$a_1 = 2$	$m_1 = 3$		$M_1 = 77$	$M_1^{-1} = 2$	308
$a_2 = 3$	$m_2 = 7$	$M = 231$	$M_2 = 33$	$M_2^{-1} = 3$	297
$a_3 = 5$	$m_3 = 11$		$M_3 = 21$	$M_3^{-1} = 10$	1050

$\frac{23\times 3}{69}$ $\frac{289}{23} =$

$77 M_1^{-1} \equiv 1 \pmod 3$ $33 M_2^{-1} \equiv 1 \pmod 7$
$M_1^{-1} = 2$ $M_2^{-1} = 3$

$21 M_3^{-1} \equiv 1 \pmod{11} \Rightarrow M_3^{-1} = 10$

$X \equiv 1655 \pmod{231}$
$X \equiv 38 \pmod{231}$

* Fermat's little theorem:-

Let p be a prime and a be any integer s.t $(a, p) = 1$ then $a^{p-1} \equiv 1 \pmod p$

34) iv) $a = 3$, $p = 23$ $286 = 22\times 13 + 3$

$(3, 23) = 1$

From FLT, $a^{p-1} \equiv 1 \pmod p$

$a^{22} \equiv 1 \pmod{23}$

$(3^{22})^{13} \equiv 1^{13} \pmod{23}$

$3^{286} \equiv 1 \pmod{23}$

$3^{286} \cdot 3^3 \equiv 3^3 \cdot 1 \pmod{23}$

$3^{289} \equiv 27 \pmod{23}$

$\Rightarrow 3^{289} \equiv 4 \pmod{23}$

Remainder $= 4$.

(ii) 3^{247} / 17

$a = 3$, $p = 17$

$247 = 16\times 15 + 7$

$3^{16} \equiv 1 \pmod{17}$ remainder

$(3^{16})^{15} \equiv 1^{15} \pmod{17}$

$\Rightarrow 3^{240} \cdot 3^7 \equiv 3^7 \pmod{17} \Rightarrow 3^{247} \equiv 11 \pmod{17}$

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(iii) $a=3,\ p=17$. $181 = 16\times 11+5$.

$3^{16} \equiv 1 \pmod{17}$

$(3^{16})^{11} \cdot 3^5 \equiv 3^5 \pmod{17}$.

$3^{181} \equiv 5 \pmod{17}$.

remainder $= 5$.

(i) $(16,7)=1 \Rightarrow 16^6 \equiv 1 \pmod 7$.

$(16^6)^8 \cdot 16^5 \equiv 16^5 \pmod 7$.

$16^{53} \equiv 4 \pmod 7$.

remainder $= 4$.

35)(i) $12x \equiv 6 \pmod 7$.

$(12,7) = 1 \mid 6$.

Eqn can be written as $5x \equiv 6 \pmod 7$.

From FLT, $a=5,\ p=7$

$5^6 \equiv 1 \pmod 7$,

$5 \cdot 5^5 \equiv 1 \pmod 7$.

Check for a multiple of 5 and leaves remainder 1 with 7.

$\Rightarrow 5\cdot 5^5 \equiv 15 \pmod 7$

$\Rightarrow 5\cdot 5^5 \equiv 5\cdot 3 \pmod 7$

By cancellation law, $5^5 \equiv 3 \pmod 7$

$12x \times 3 \equiv 6\times 3 \pmod 7$.

$36x \equiv 18 \pmod 7$

$\Rightarrow x \equiv 4 \pmod 7$

(OR)

$x \equiv a^{p-2} b \pmod p$
$x \equiv 5^{7-2}\, 6 \pmod 7 \Rightarrow x \equiv 3 \cdot 6 \pmod 7$.
$\Rightarrow x \equiv 4 \pmod 7$.

(ii) $37x \equiv 5 \pmod{11}$.
$\Rightarrow 4x \equiv 5 \pmod{11}$.
$\Rightarrow a = 4,\ p = 11$.
$\Rightarrow 4^{10} \equiv 1 \pmod{11}$.
$\Rightarrow 4 \cdot 4^9 \equiv 1 \pmod{11}$.
$\Rightarrow 4 \cdot 4^9 \equiv 4 \cdot 3 \pmod{11}$.
$\Rightarrow 4^9 \equiv 3 \pmod{11}$.
$\Rightarrow x \equiv 4^{11-2}\, 5 \pmod{11}$
$\Rightarrow x \equiv 3 \cdot 5 \pmod{11}$
$\Rightarrow x \equiv 4 \pmod{11}$

(iv) $3x \equiv 4 \pmod 5$.
$\Rightarrow a = 3,\ p = 5$.
$\Rightarrow 3^4 \equiv 1 \pmod 5$
$\Rightarrow 3 \cdot 3^3 \equiv 1 \pmod 5$
$\Rightarrow 3 \cdot 3^3 \equiv 3 \cdot 2 \pmod 5$
$\Rightarrow 3^3 \equiv 2 \pmod 5$
$\Rightarrow x \equiv 3^{5-2} \cdot 4 \pmod 5$
$\Rightarrow x \equiv 2 \cdot 4 \pmod 5$
$\Rightarrow x \equiv 3 \pmod 5$

* Euler's Totient Function / Euler's ϕ function:

Denoted by $\phi(n)$ and is defined as the no. of ~~possi~~ positive integers less than n and relatively prime with n.
Eg: $n = 6 \Rightarrow \phi(6) = 2$
$n = 12 \Rightarrow \phi(12) = 4$
↳ 1, 5, 7, 11.

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In general if p is prime, $\phi(p) = p-1$

1) If p, q are co-prime then $\cancel{\phi(pq)}$

$$\phi(pq) = \phi(p)\phi(q) = (p-1)(q-1).$$

2) $\phi(pq) = (p-1)(q-1)$.

39). $n = pq = 19939$

$$\phi(pq) = (p-1)(q-1)$$
$$= pq - (p+q) + 1$$
$$19656 = 19939 - (p+q) + 1$$
$$\Rightarrow \quad p+q = 284.$$
$$pq = 19939.$$
$$\Rightarrow \frac{19939}{q} + q - 284 = 0$$
$$\Rightarrow q^2 - 284q + 19939 = 0.$$

$p = 157,\ q = 127$ or.
$q = 157,\ p = 127$.